

Randomized and low-rank approximation

Convexity of the expected error of volume-sampled column subsets

Difficulty: challenging **Importance:** interesting to the community

Status: Lean verified

Rating rationale: Challenging because the required second-difference inequality goes beyond standard log-concavity; community impact is understanding average column-selection and Nystrom error.

Lean proof and verification evidence — 2026-09-12

The complete original discrete-convexity assertion is Lean verified. The [proof at revision bf144a8](#) covers every $n \geq 3$, strictly positive real tuple and $2 \leq j \leq n - 1$, including repeated eigenvalues and both endpoint indices, without ordering or factorization assumptions.

Mathematical proof: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Lean formalization:** George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, with AI-agent assistance.

The actual generating-polynomial derivative of order $j - 1$ has $n - j + 1 \geq 2$ positive reciprocal-root factors μ_a , with multiplicity. The proof derives

$$f(j-1) - 2f(j) + f(j+1) = \frac{2 \sum_{a < b} \mu_a \mu_b (\mu_a - \mu_b)^2}{s_1(s_1^2 - s_2)} \geq 0, \quad s_r = \sum_a \mu_a^r,$$

including strict positivity of the denominator. The six [checked exports](#), each with prefix *NLA.RA07.*, are:

- *elementary_values*: actual subset-sum conventions and positivity.
- *generating_derivative_values*: actual coefficients and factorial-scaled derivatives.
- *positive_derivative_factorization*: exact degree, real splitting and every root multiplicity.
- *power_sum_certificate*: positive denominator and exact nonnegative pair identity.
- *second_difference_certificate*: that identity for every original ratio and index.
- *errorSequence_convex*: the complete affirmative canonical theorem.

Separate pairs of independent agents approved the [statements and proof](#). [Linux run 34715563781](#) matched all six exports with the sandboxed Comparator and replayed the solution in Lean's default kernel. The [operational audit and original artifacts](#) bind 123 inputs and the isolation/rejection controls. Twelve transitive axiom checks use only *propext*, *Classical.choice* and *Quot.sound*. No external human peer review is claimed.

The pins are [Lean 4.33.1](#), [LeanCert 621a43d](#) and [Mathlib 0df444a](#). LeanCert audits kernel trust; there is no numerical interval certificate or approximate root computation. See the [manifest](#), [dependency pins](#) and [exact targets](#). From the verified revision on a [non-root Linux host](#):

```
tools/lean/bootstrap.sh /absolute/path/to/nla-lean-tools
tools/lean/selftest.sh /absolute/path/to/nla-lean-tools
tools/lean/verify.sh \
  randomized-and-low-rank-approximation/RA-07/lean \
  /absolute/path/to/nla-lean-tools
```

The source's additional monotonicity, index-one, sampling-expectation and application claims are outside these six exports. The complete original scalar convexity target and its historical informal proof remain below.

Resolution — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the exact target.

The sequence $(j+1)e_{j+1}/e_j$ is decreasing and discretely convex for every positive spectrum, including both endpoints. The exact second-difference certificate proves the full canonical conjecture.

The complete target is resolved. Its former difficulty rating is historical; the original mathematical statement and source evidence are retained below.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1.1**. [Independent proof review](#) · [Authorship and submission record](#). The 2026-09-11 verification was independent agent review, without external human peer review or formal certification. The later Lean verification above covers the complete canonical scalar convexity target.

For $n \geq 3$ and positive real numbers $\lambda_1, \dots, \lambda_n$, let

$$e_j(\lambda) = \sum_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=j}} \prod_{i \in S} \lambda_i, \quad e_0 = 1, \quad e_{n+1} = 0.$$

Define $f(j) = (j+1)e_{j+1}(\lambda)/e_j(\lambda)$ for $1 \leq j \leq n$. Is f discretely convex, that is,

$$f(j-1) - 2f(j) + f(j+1) \geq 0 \quad (2 \leq j \leq n-1),$$

for every such n and positive tuple λ ?

For a matrix A with positive eigenvalues λ_i of $A^T A$, sample a set S of j columns with probability proportional to $\det(A_S^T A_S)$. Its expected squared Frobenius projection error is

$$\mathbb{E} \| (I - P_{A_S}) A \|_F^2 = f(j),$$

where P_{A_S} is the orthogonal projector onto the selected column span. Thus the conjecture asserts a diminishing marginal improvement in the average reconstruction error of these fixed-size determinantal samples. It is distinct from optimizing a worst-case approximation factor over all matrices with a prescribed spectrum.

References

1. M. Dereziński, R. Khanna, and M. W. Mahoney, *Improved guarantees and a multiple-descent curve for Column Subset Selection and the Nyström method*, NeurIPS 2020, arXiv:2002.09073. Section 5, Conjecture 1 is exactly the discrete-convexity statement. [Paper](#).
2. A. Deshpande, L. Rademacher, S. S. Vempala, and G. Wang, *Matrix Approximation and Projective Clustering via Volume Sampling*, Theory of Computing 2 (2006), Article 12, pp. 225–247. See the volume-sampling matrix approximation results. [Paper](#).

Status check — 2026-09-08

Searched “Dereziński Khanna Mahoney Conjecture 1 convexity”, “elementary symmetric ratio $k+1$ convexity DPP”, and the exact 2020 paper title with 2025/2026. Checked the current arXiv record. No proof or counterexample was found. Standard Newton inequalities imply other ratio inequalities but are not cited as a resolution of this second-difference inequality.

Audit — 2026-09-10

Rechecked [Dereziński–Khanna–Mahoney, §5, Conjecture 1](#); the current record remains v3 of December 2020. Volume-sampling convexity and elementary-symmetric-ratio searches found no resolution. This remains historical explicit conjecture evidence, supplemented by a bounded follow-up search.